

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Cell: The Unit of Life

LECTURE-7

ZOOLOGY

DAVOOD SIR (CEO of NG)



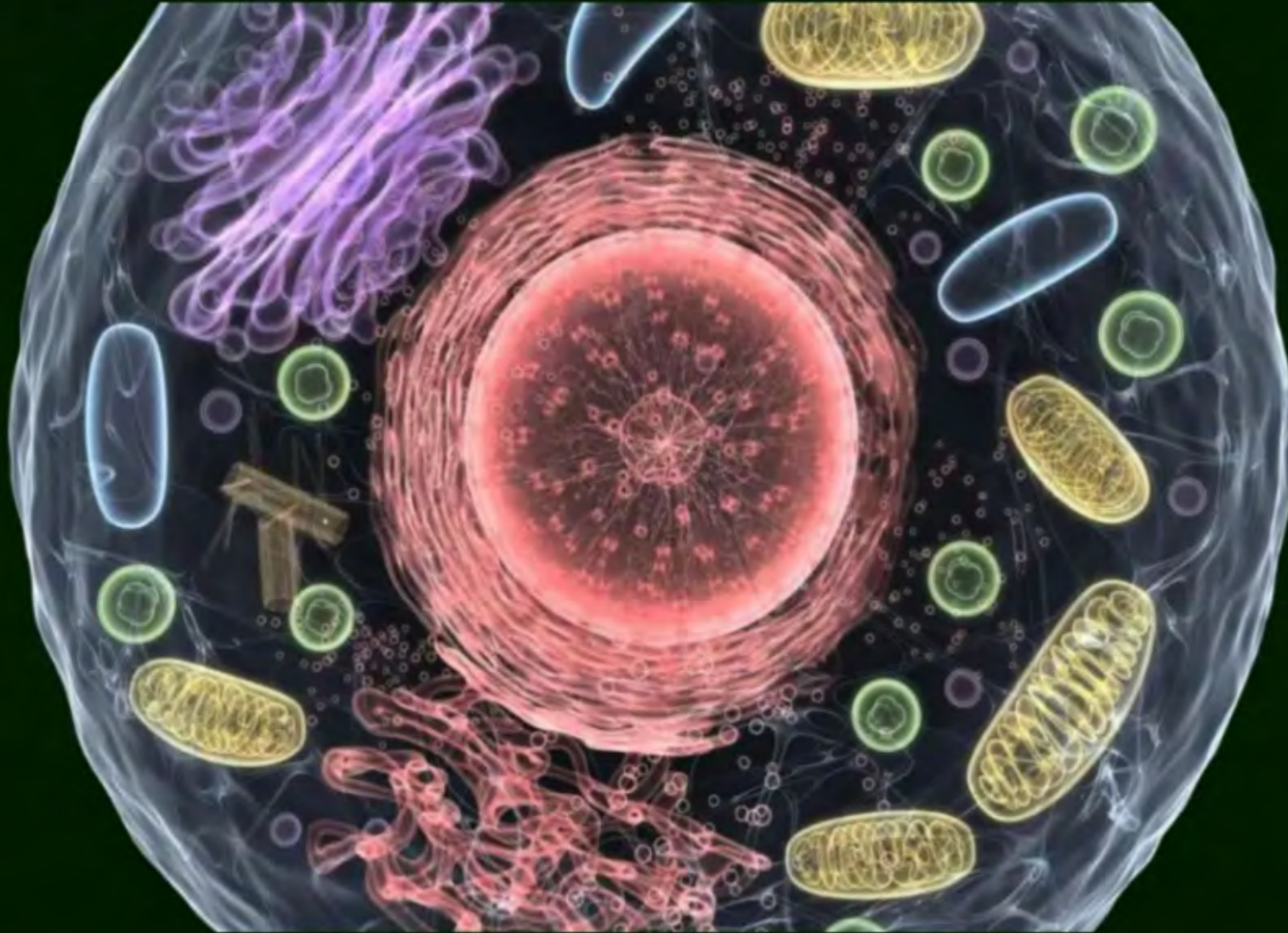
Topics To Be Covered

- 1 Cytoskeleton
- 2 Microbodies
- 3 Nucleus
- 4 Previous Year NEET Questions.

Eukaryotic Cell



Cytoskeleton



Cytoskeleton

An elaborate network of filamentous proteinaceous structures consisting of **microtubules, microfilaments and intermediate filaments** present in the cytoplasm is collectively referred to as the cytoskeleton.

Functions:

- Such as mechanical support, motility, maintenance of the shape of the cell.

Question: NEET-2023**Kon Banayga Nabkaar**

Which of the following functions is carried out by cytoskeleton in a cell?

1 Motility ✓

2 Transportation ✗

3 Nuclear division ✗

4 Protein synthesis ✗



Microbodies

They are small single membrane bound minute vesicles or cell organelles that contain various **enzymes** and are present in both plant and animal cells. Examples are Peroxisomes and Glyoxysomes.

FUNCTION:-

- Their primary function includes the breakdown of fats, alcohols, and amino acids.
- In animal cells, they metabolise in number of toxic substances.



Nucleus

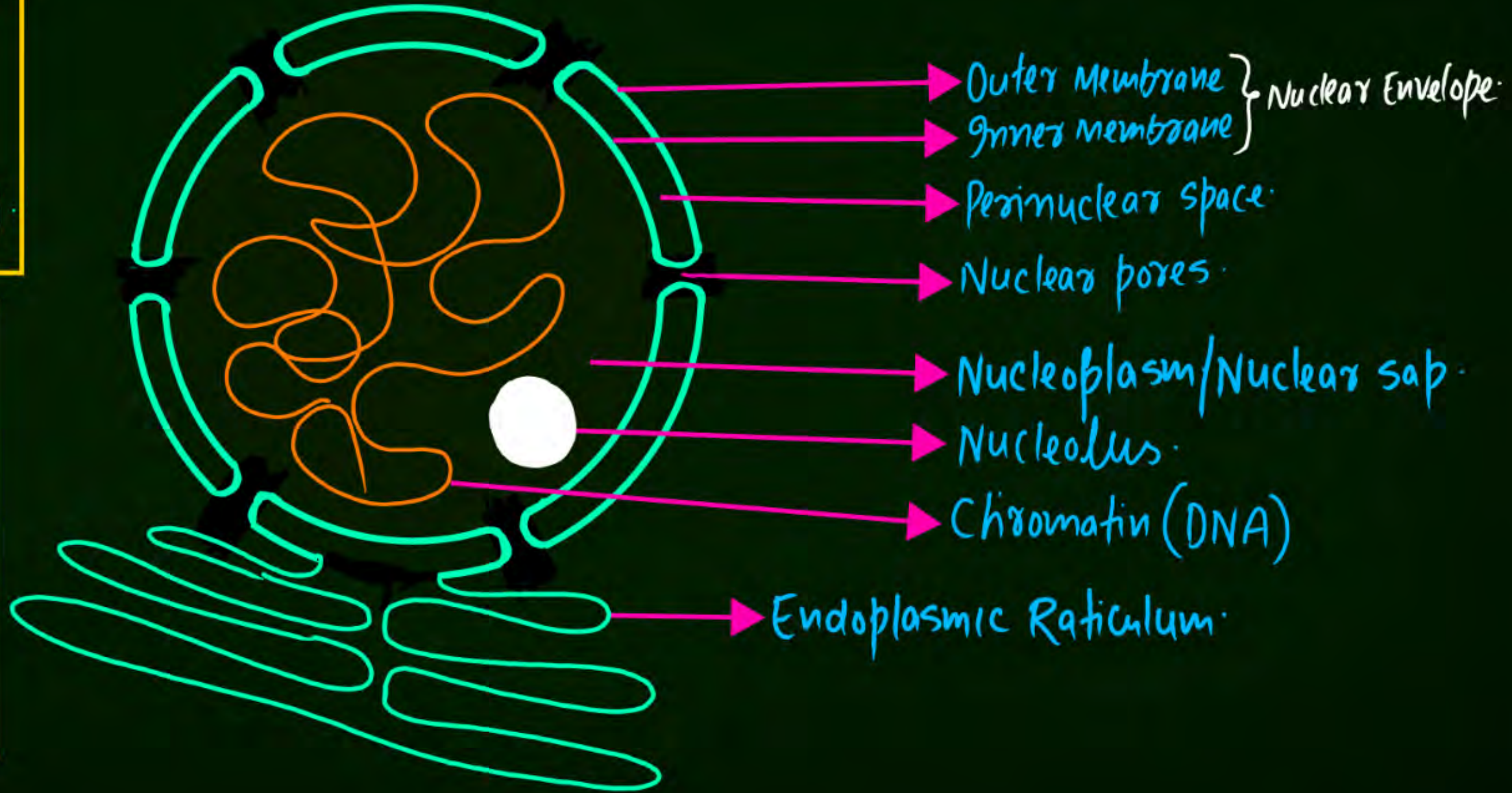
→ Boss of the cell
→ Discovered by Robert Brown.

Anucleated cell

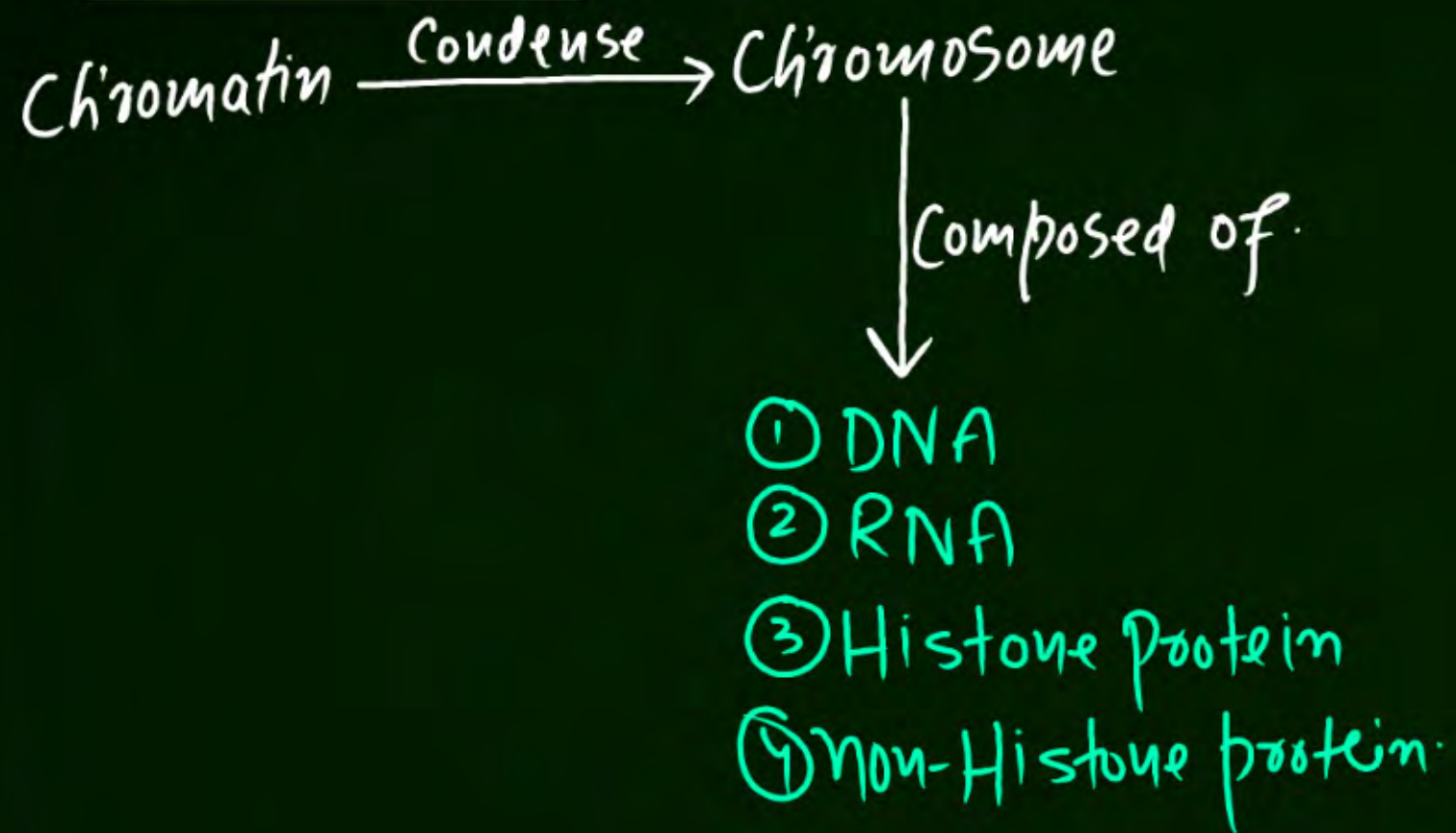
- RBC (mammals)
- Sieve tube cell (phloem)

Nucleolus

- ① Membraneless.
- ② RNA Synthesis

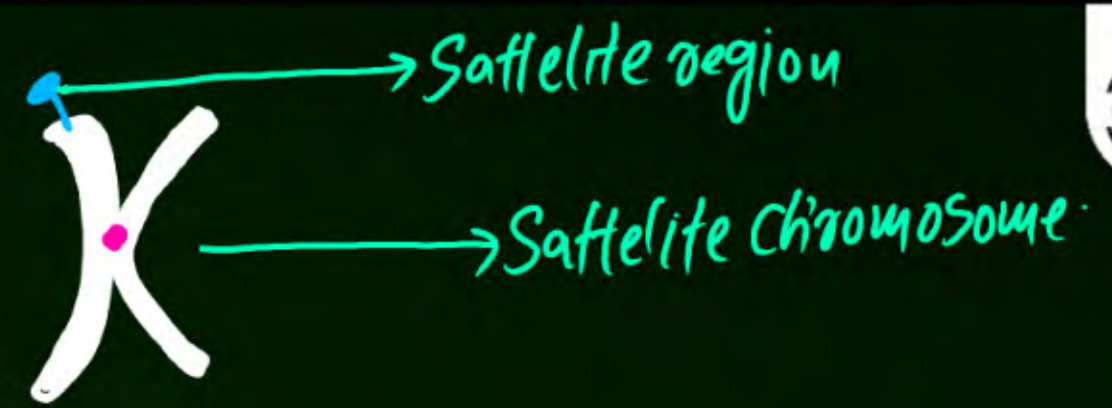


Nucleus

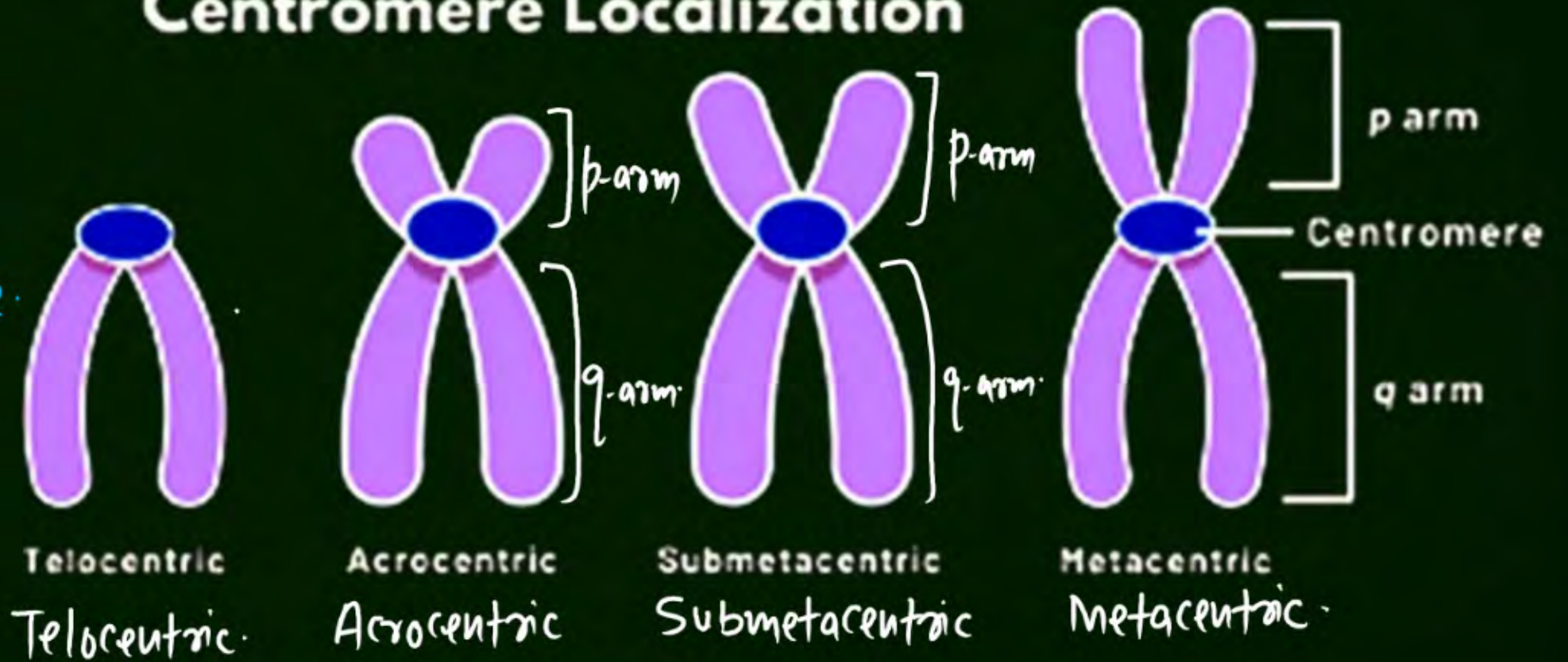
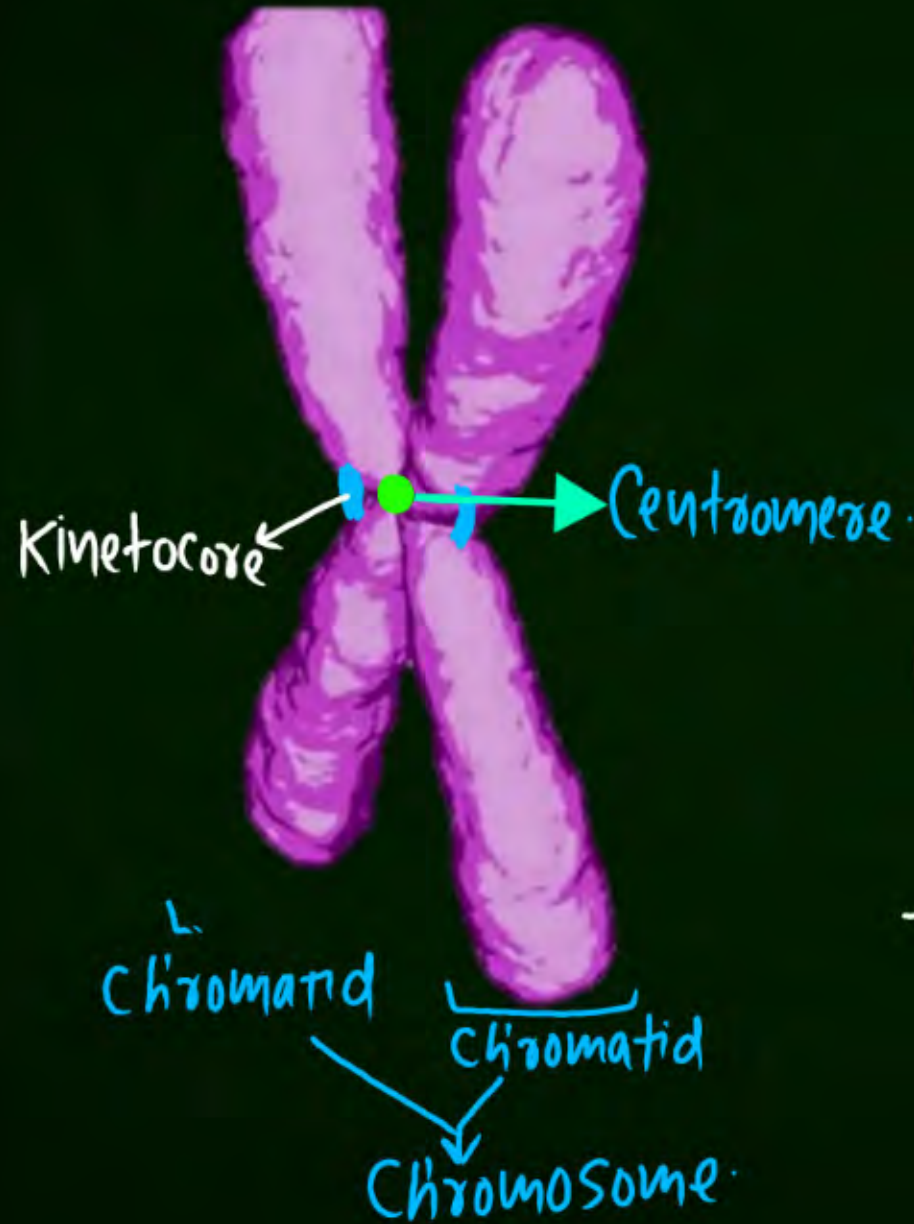


Nucleus

→ Chromosome



Centromere Localization



Nucleus

First described by Robert Brown (1831) & it Contains:-

- a) *Nuclear matrix.*
- b) *Chromatin (highly extended thread-like).*
- c) *One or more nucleoli*
- **Nuclear Envelope:-** Discovered clearly by electron microscopy & is double membrane structure. Space between membranes is called Perinuclear space. It acts as barrier between nucleus and cytoplasm. Outer membrane is continuous with endoplasmic reticulum (ER) which bears ribosomes.
- **Nuclear Pores:-** Formed by fusion of inner and outer nuclear membranes. They allow bidirectional transport of RNA & Proteins.
- **Chromatin:-** Stains with basic dyes & was shown by Flemming.
- **Nucleolus:-** Non-membrane bound, Spherical, present in nucleoplasm & is the Site of rRNA synthesis. They are large and numerous in cells with high protein synthesis.

Nucleus

- Cells Lacking Nucleus are:-
 - a) Mature mammalian erythrocytes (RBCs),
 - b) Sieve tube cells of vascular plants.

Chromatin & Chromosomes

Nucleus contains a loose, indistinct network of nucleoprotein fibres called chromatin. During cell division, chromatin condenses to form distinct chromosomes. Chromatin is composed of:

- a) DNA, b) Histone proteins (basic), c) Non-histone proteins, d) RNA

Chromosome Structure:- Chromosomes are visible only during dividing stages. Each chromosome has a primary constriction called the centromere. Centromere holds two sister chromatids together. On both sides of the centromere are disc-shaped kinetochores.

Types of Chromosomes (Based on Centromere Position):-

- a) **Metacentric:-** Centromere in the middle and is having two equal arms.
- b) **Sub-metacentric:-** Centromere slightly away from middle & forms one short & one long arm.
- c) **Acrocentric:-** Centromere near one end & forms one very short arm and one very long arm.
- d) **Telocentric:-** Centromere at the terminal end.

Secondary Constriction & Satellite:- Some chromosomes show secondary constrictions that do not stain. These form a small detached-looking part called a satellite. Such chromosomes are called satellite chromosomes.

NEET FOCUS POINTS

1. **Perinuclear space.** ✓
2. **Nuclear pores → RNA & protein transport.**
3. **rRNA synthesis → nucleolus**
4. **RBCs & sieve tubes → enucleate**
5. **Chromatin → DNA + histones**
6. **Centromere → holds chromatids**
7. **Kinetochores → spindle attachment**
8. **Satellite → secondary constriction**

Question: NEET-2024

Kon Banayga Nabkaar

Match List-I with List-II.

Column-I

Column-II

- | | |
|----------------------|------------------------|
| A. Axoneme | I. Centriole |
| B. Cartwheel pattern | II. Cilia and flagella |
| C. Crista | III. Chromosome |
| D. Satellite | IV. Mitochondria |

Choose the correct answer from the options given below

1 A-(IV), B-(III), C-(II), D-(I) ✗

2 A-(IV), B-(II), C-(III), D-(I) ✗

3 A-(II), B-(IV), C-(I), D-(III) ✗

4 A-(II), B-(I), C-(IV), D-(III) ✓

Question: NEET-2024

Kon Banayga Nabkaar

Match List-I with List-II.

Column-I

A. Nucleolus

B. Centriole

C. Leucoplasts

D. Golgi apparatus

Column-II

I. Site of formation of glycolipid

II. Organization like the cartwheel

III. Site for active ribosomal RNA synthesis

IV. For storing nutrients

Choose the correct answer from the options given below

1 A-(III), B-(II), C-(IV), D-(I)

2 A-(II), B-(III), C-(I), D-(IV)

3 A-(III), B-(IV), C-(II), D-(I)

4 A-(I), B-(II), C-(III), D-(IV)

Question: NEET-2022

Kon Banayga Nabkaar

Match List-I with List-II.
Column-I

A. Metacentric chromosome

B. Acrocentric chromosome

C. Sub-metacentric

D. Telocentric chromosome

I.

II.

III.

IV.

Centromere situated close to the end forming one extremely short and one very long arm

Centromere at the terminal end

Centromere in the middle forming two equal arms of chromosomes

Centromere slightly away from the middle forming one shorter arm and one longer arm.

Column-II

Choose the correct answer from the options given below

1

A-(III), B-(I), C-(IV), D-(II)

2

A-(I), B-(III), C-(II), D-(IV)

3

A-(II), B-(III), C-(IV), D-(I)

4

A-(I), B-(II), C-(III), D-(IV)

Question: NEET-2021

Kon Banayga Nabkaar

Which of the following is an incorrect statement?

- 1 Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
- 2 Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
- 3 Microbodies are present both in plant and animal cells.
- 4 The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.

Question: NEET-2021

Kon Banayga Nabkaar

When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as

1 Acrocentric

2 Metacentric

3 Telocentric

4 Sub-metacentric.



Thank
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